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(54) **REFILL SYSTEM AND COSMETIC PACKAGE INCORPORATING SAME**

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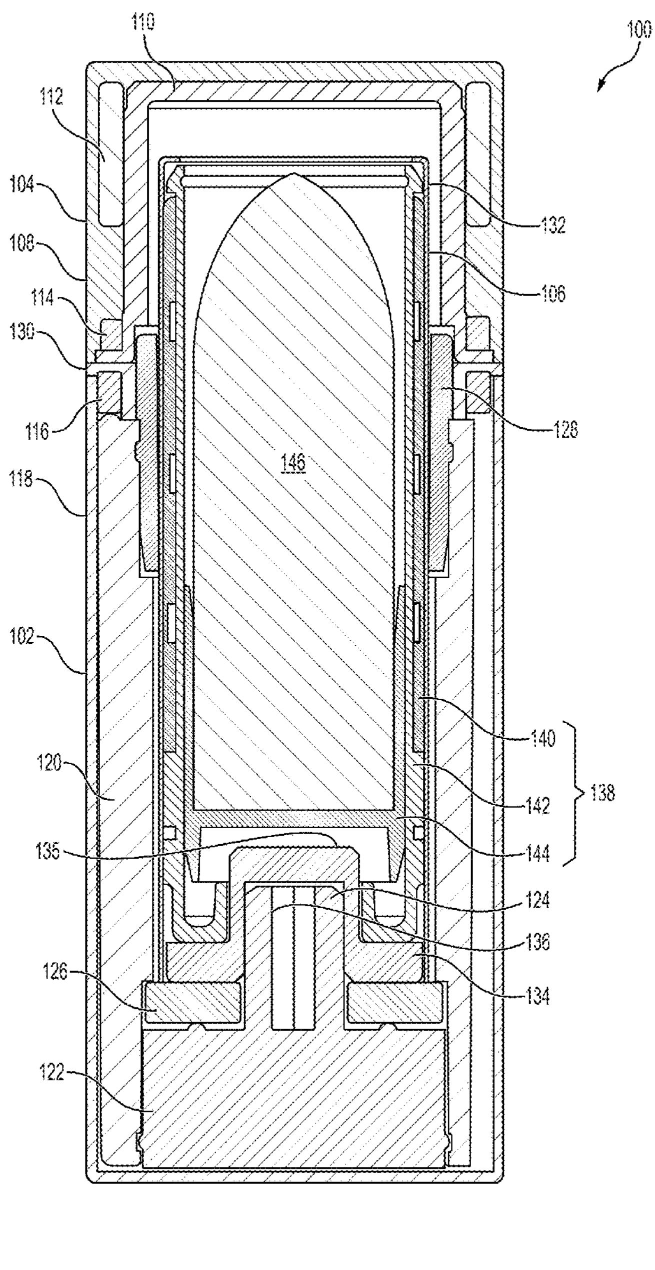
(57) **ABSTRACT**

Refill systems and cosmetic packages having such refill systems are provided herein. In some embodiments, a cosmetic package includes: a base having an inner volume and a drive post disposed proximate a bottom of the base, the drive post extending into the inner volume; a refill cartridge configured to be removably disposed within the inner volume of the base, the refill cartridge including a recess configured to mate with the drive post; and a cap configured to rest on the base and enclose the refill cartridge within the inner volume.



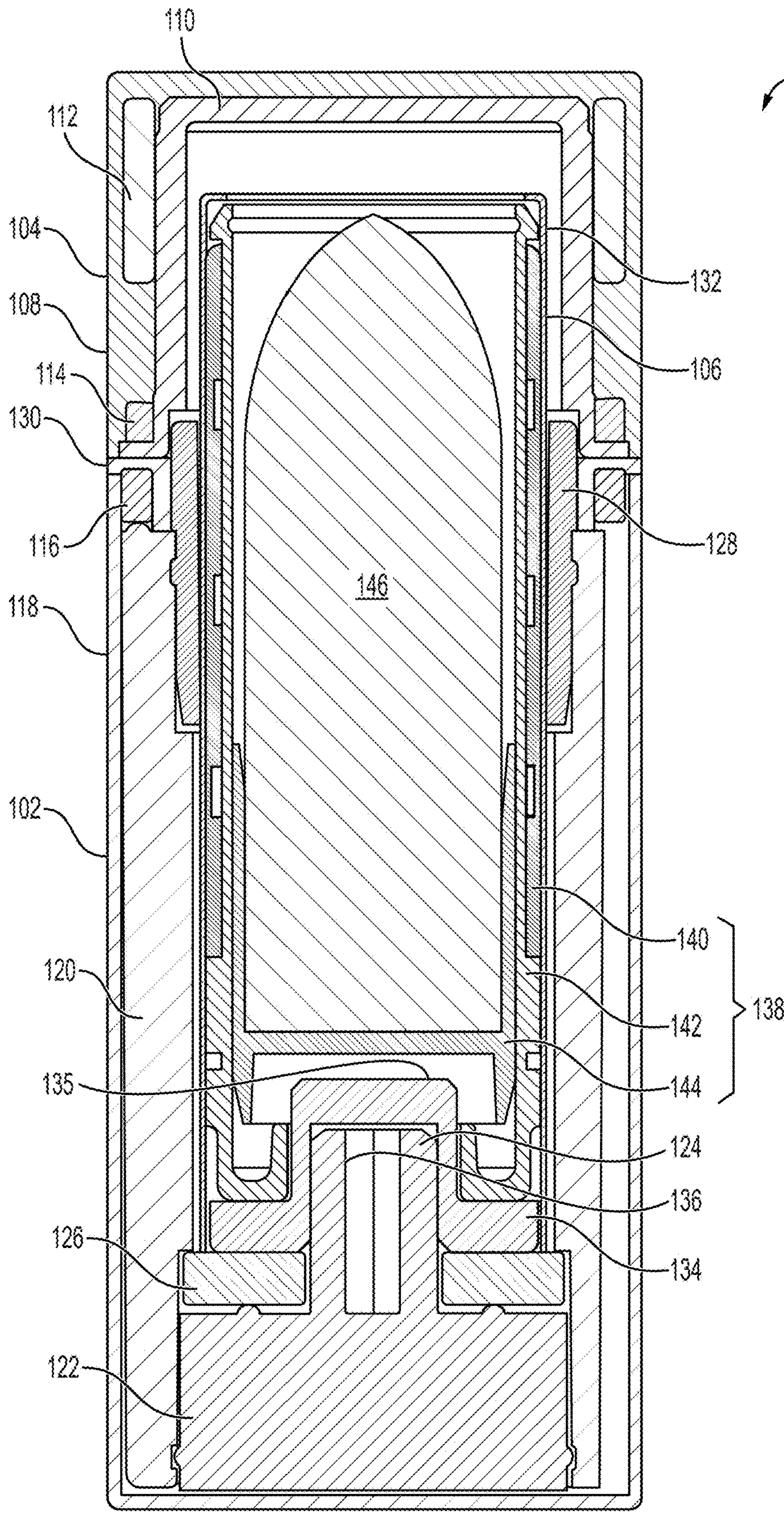

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FIG. 1

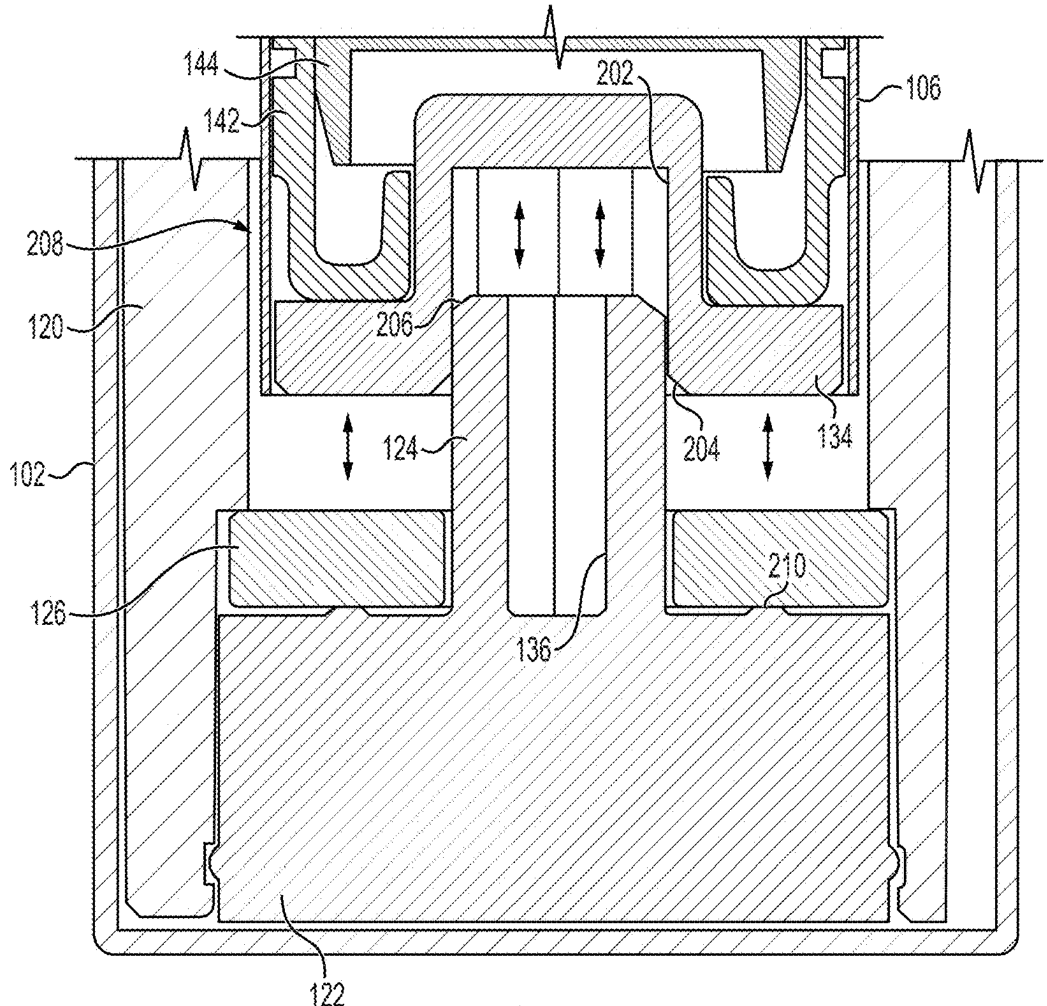
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FIG. 2

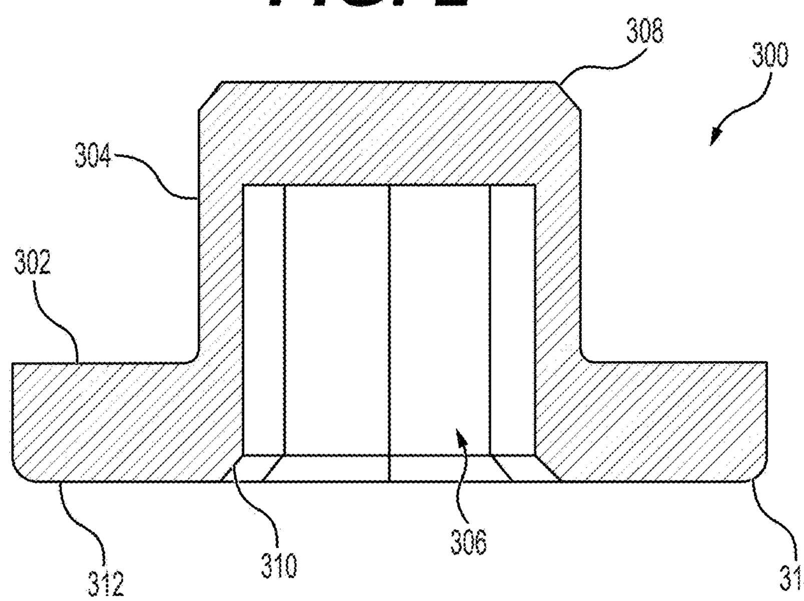
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FIG. 3

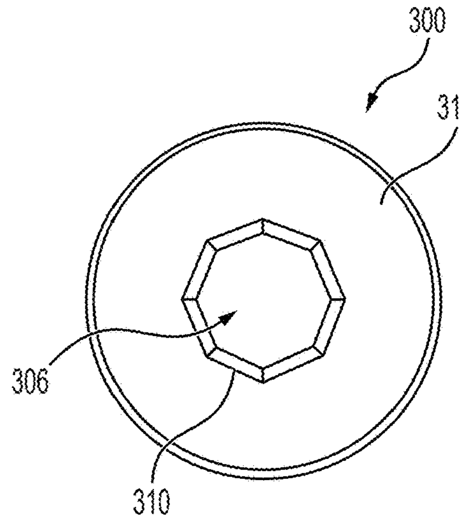

FIG. 4

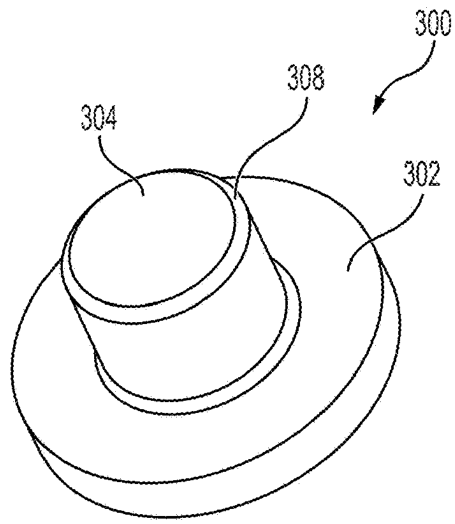

FIG. 5

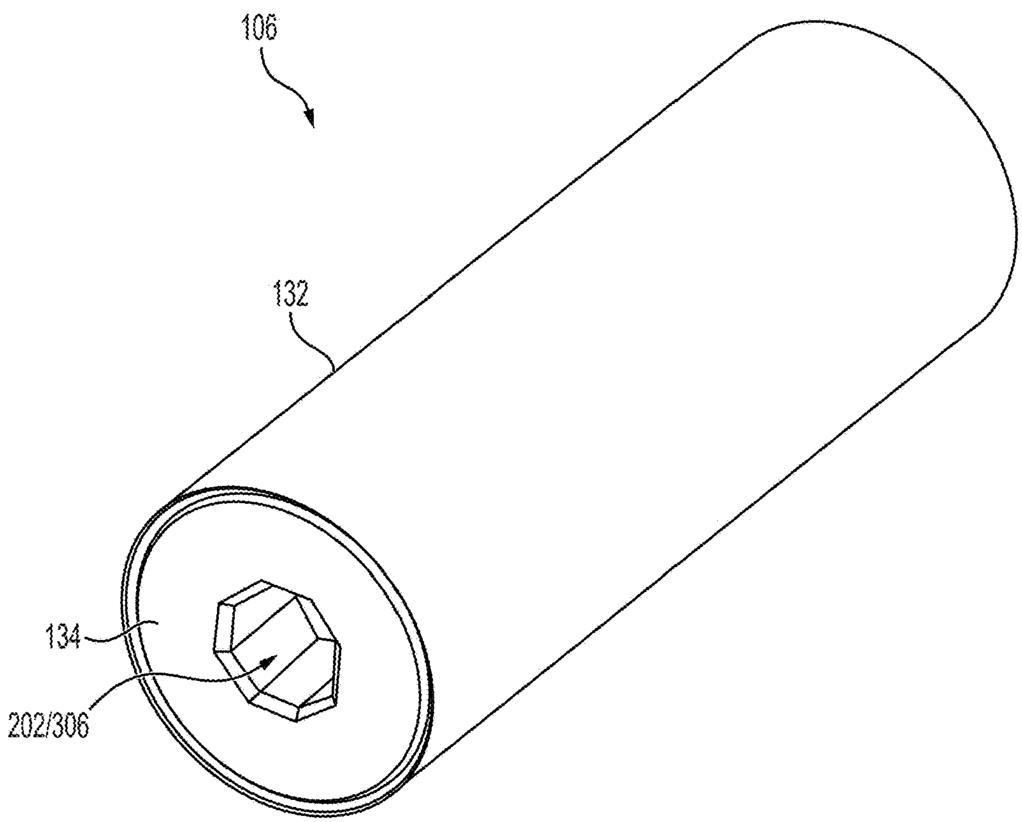

FIG. 6

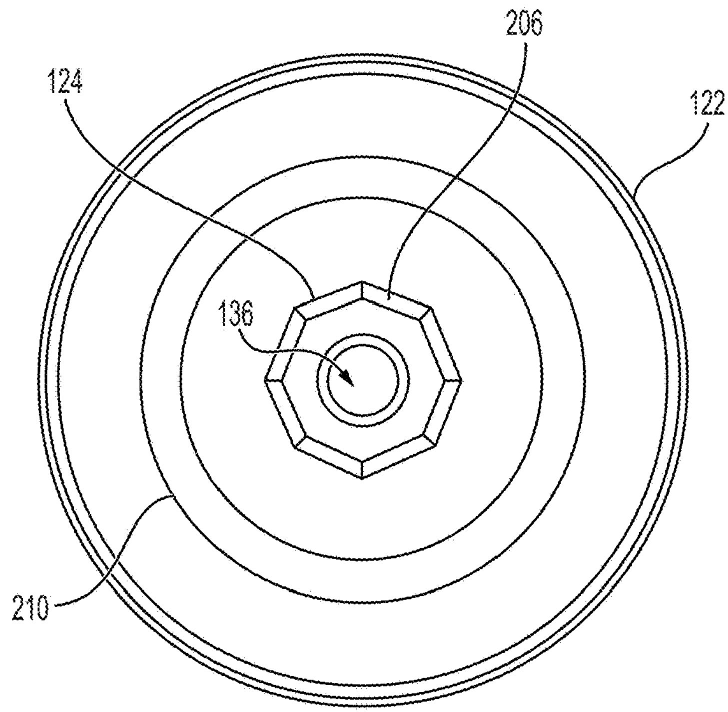

FIG. 7

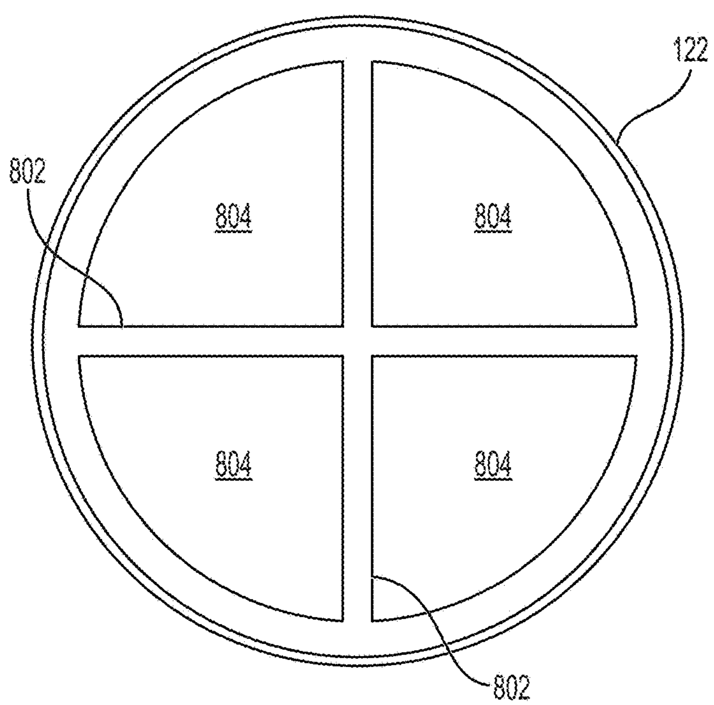

FIG. 8

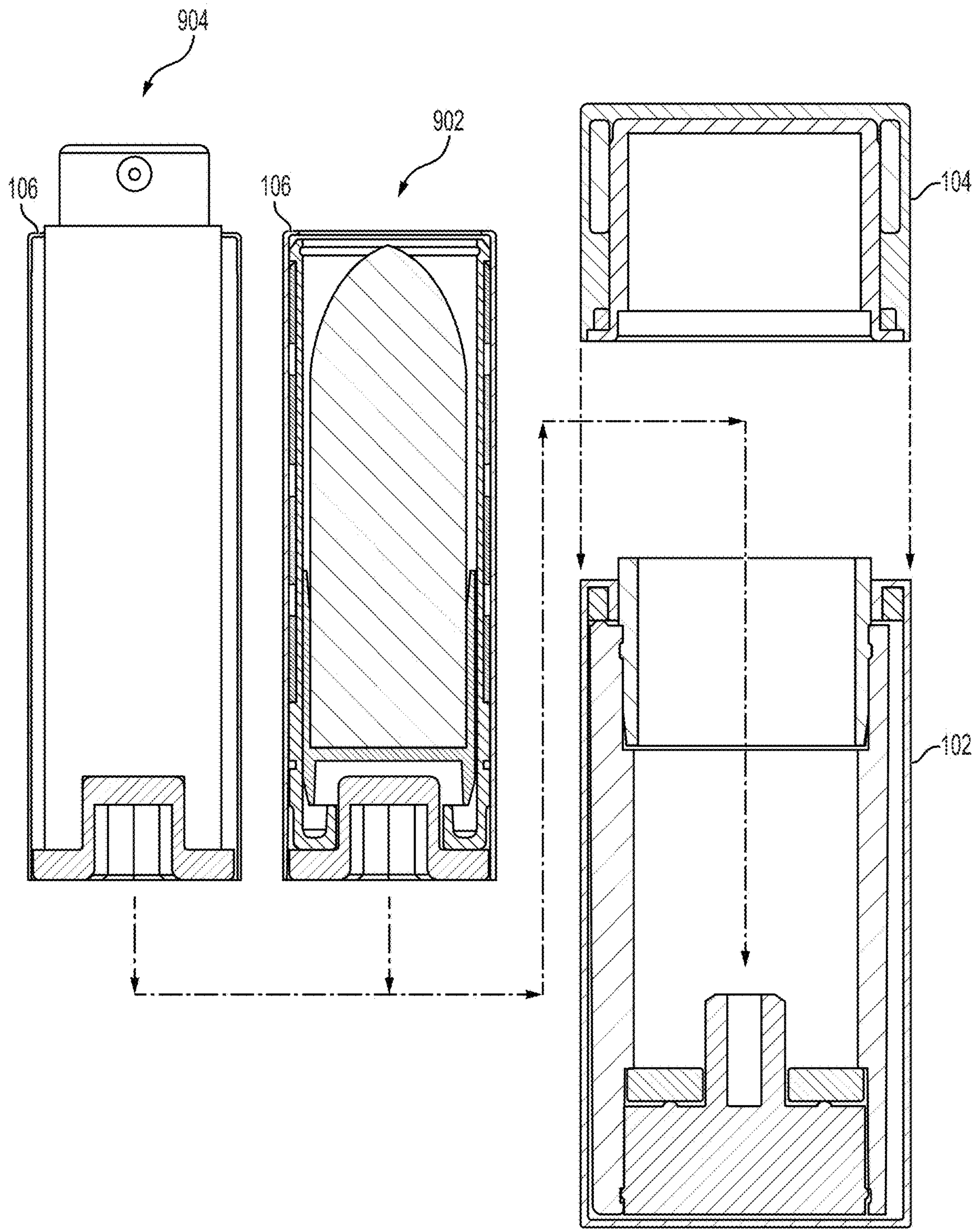

FIG. 9

REFILL SYSTEM AND COSMETIC PACKAGE INCORPORATING SAME

FIELD

[0001] Embodiments of the present disclosure generally relate to refillable packages for cosmetics.

BACKGROUND

[0002] Most cosmetic packages, or containers, are intended for single use by a consumer and are disposable. The inventors have provided an improved refill system and cosmetic packages incorporating such refill systems.

SUMMARY

[0003] Refill systems and cosmetic packages having such refill systems are provided herein. In some embodiments, a cosmetic package includes: a base having an inner volume and a drive post disposed proximate a bottom of the base, the drive post extending into the inner volume; a refill cartridge configured to be removably disposed within the inner volume of the base, the refill cartridge including a recess configured to mate with the drive post; and a cap configured to rest on the base and enclose the refill cartridge within the inner volume.

[0004] In some embodiments, a cosmetic package includes: a base having an inner volume and a bottom plug disposed proximate a bottom of the base, the bottom plug including a drive post extending into the inner volume; an annular magnet disposed atop the bottom plug and having the drive post extending through the central opening of the annular magnet; a refill cartridge configured to be removably disposed within the inner volume of the base, the refill cartridge having a bottom formed of a magnetic material and including a recess configured to mate with the drive post; and a cap configured to rest on the base and enclose the refill cartridge within the inner volume.

[0005] In some embodiments, a cosmetic package includes: a base having an inner volume and a bottom plug disposed proximate a bottom of the base, the bottom plug including a drive post extending into the inner volume, wherein the drive post is octagonal and includes a chamfer along an upper edge of the drive post; an annular magnet disposed atop the bottom plug and having the drive post extending through the central opening of the annular magnet; a refill cartridge configured to be removably disposed within the inner volume of the base, the refill cartridge having a bottom formed of a magnetic material and including an octagonal recess configured to mate with the drive post, the recess having a chamfer disposed along a lower edge of the recess to facilitate alignment of the recess with the drive post; and a cap configured to rest on the base and enclose the refill cartridge within the inner volume, wherein the cap and the base are configured to be magnetically retained to each other.

[0006] Other and further embodiments of the present disclosure are described below.

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] Embodiments of the present disclosure, briefly summarized above and discussed in greater detail below, can be understood by reference to the illustrative embodiments of the disclosure depicted in the appended drawings. However, the appended drawings illustrate only typical embodiments of the disclosure and are therefore not to be considered limiting of scope, for the disclosure may admit to other equally effective embodiments.

[0008] FIG. **1** is a schematic, cross-sectional side view of a cosmetic package having a refill system in accordance with embodiments of the present disclosure.

[0009] FIG. **2** is partial, close-up view of a bottom portion of the cosmetic package of FIG. **1**.

[0010] FIG. **3** is a cross sectional side view of a bottom portion of a refill cartridge for a cosmetic package having a refill system in accordance with embodiments of the present disclosure.

[0011] FIG. **4** is a bottom view of a portion of a base of a refill cartridge of a cosmetic package having a refill system in accordance with embodiments of the present disclosure.

[0012] FIG. **5** is an isometric top view of a portion of a base of a refill cartridge of a cosmetic package having a refill system in accordance with embodiments of the present disclosure.

[0013] FIG. **6** is an isometric bottom view of a refill cartridge for a cosmetic package having a refill system in accordance with embodiments of the present disclosure.

[0014] FIG. **7** is a top view of a base of a cosmetic package having a refill system in accordance with embodiments of the present disclosure.

[0015] FIG. **8** is a bottom view of a base of a cosmetic package having a refill system in accordance with embodiments of the present disclosure.

[0016] FIG. **9** is an exploded schematic side view showing interrelation of major components of a cosmetic package having a refill system in accordance with embodiments of the present disclosure.

[0017] To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the figures. The figures are not drawn to scale and may be simplified for clarity. Elements and features of one embodiment may be beneficially incorporated in other embodiments without further recitation.

DETAILED DESCRIPTION

[0018] Embodiments of refill systems and cosmetic packages having such refill systems are provided herein. Although the refill system and cosmetic package described herein illustratively holds lipstick, the inventive refill systems and cosmetic packages can be used to store and apply any liquid or solid cosmetics, including fragrances. The inventive refill systems and cosmetic packages disclosed herein advantageously provide a convenient refillable package having a sleek, attractive, and easy to use refill cartridge.

[0019] FIG. **1** is a schematic, cross-sectional side view of a cosmetic package **100** having a refill system in accordance with embodiments of the present disclosure. The cosmetic package **100** generally includes a cylindrical base **102**, cap **104**, and refill cartridge **106**. The refill cartridge **106** is disposed within an opening of the base **102** and removably coupled to the base **102** such that the refill cartridge **106** can be completely removed and replaced after refilling or with a different refill cartridge. The cap **104** is configured to fit over a portion of the refill cartridge **106** that protrudes above the base **102** and to rest on the base **102** to protect the refill cartridge **106** (and any product contained therein).

[0020] In some embodiments, the cap **104** includes an outer cap **108** and an inner cap **110**. A cap weight **112** can be disposed between the outer cap **108** and the inner cap **110** to provide improved feel and stability in the cap **104**. The cap weight can be any suitable material with sufficient mass to provide a desired weight given the size of the cap weight **112**, such as iron, steel, or the like. In some embodiments, a magnet **114** can be disposed proximate to a bottom portion of the cap **104** to facilitate securely retaining the cap **104** on the base **102**, as discussed in more detail below. For example, the magnet **114** can be disposed between the outer cap **108** and the inner cap **110** to more securely retain the magnet **114** within the cap **104** and/or to hide the magnet **114** from view. The magnet **114** can be any suitable magnet having a magnetic strength that is strong enough to retain the cap **104** when disposed on the base **102** and that is weak enough to facilitate manual removal of the cap **104** by a user. For example, in some embodiments, a separation force between the cap **104** and the base **102** can range from 0.1 to 3 kilogram-force (kgf), such as about 0.3 to 2 kgf. In some embodiments, the magnet **114** can be a neodymium magnet, such as but not limited to an N**33** or N**38** magnet or the like. Alternatively or in combination, in some embodiments the cap **104** can be retained on the base **102** with the same separation forces by, for example, providing a snap or friction fit between the cap **104** and the base **102**.

[0021] The base **102** generally includes an outer shell **118** that surrounds the components of the base **102** and provides a sleek, unified outer appearance and/or more finished look. The outer shell **118** surrounds an annular inner shell **120** having a central opening to an inner volume configured to receive the refill cartridge **106**. The inner shell **120** can be glued or otherwise fixed to the outer shell **118**. A base plug **122** can be disposed at the bottom of the inner shell **120**. The base plug **122** can be secured in any suitable manner to the bottom of the inner shell **120** (e.g., press fit, bonded, pinned, screwed, or the like). A drive post **124** extends from base plug **122** axially along a central axis of the base **102** into the hollow central opening of the inner shell **120**. The drive post **124** interfaces with the refill cartridge **106** to control dispensing or position of product contained in the refill cartridge **106** as discussed in more detail below. The drive post **124** and the base plug **122** can be separate components coupled together or the drive post **124** and the base plug **122** can be formed from a single piece of material.

[0022] In some embodiments, the drive post **124** can include an opening **136** to reduce the weight of the base plug **122** and the amount of material used in fabricating the base plug **122**. The opening **136** can also provide a region for air flow during insertion of refill cartridge **106** to facilitate ease of insertion. FIG. **7** depicts a top view of the base plug **122** in accordance with some embodiments of the present disclosure to more clearly show one exemplary configuration of the drive post **124** and the protrusion **210** (discussed below with respect to FIG. **2**). FIG. **8** depicts a bottom view of the base plug **122**. As shown in FIG. **8**, in some embodiments, the base plug **122** can be hollow and can include one or more ribs **802** defining one or more recesses **804**. Although two perpendicular ribs **802** defining four recesses **804** are shown, other numbers of ribs can be provided at varying angles to define different numbers of recesses. Alternatively, ribs **804** can be omitted and the base plug **122** can be solid or the base plug **122** can be hollow with a single recess **804**.

[0023] Returning to FIG. **1**, a magnet **126** can be disposed atop the base plug **122**. In some embodiments, the magnet **126** can be secured mechanically between the base plug **122** and a ledge formed in the inner shell **120**, for example, by providing a counterbore into the bottom portion of the inner shell **120**. In some embodiments the magnet **126** can alternatively or additionally be glued or otherwise bonded to the base plug **122**. In some embodiments, one or more protrusions (e.g., protrusion **210** shown in FIGS. **2** and **7**) can extend from the upper surface of the base plug **122** to support the magnet **126** in a spaced apart relation to the base plug **122**, forming a gap therebetween. An adhesive or other bonding agent can be disposed in the gap to secure the magnet **126** to the base plug **122**. The magnet **126** can have a ring shape to allow the drive post **124** to extend through and beyond the magnet **126**.

[0024] The magnet **126** can be any suitable magnet having a magnetic strength that is strong enough to retain the refill cartridge **106** when disposed in the base **102** during use and handling of the cosmetic package and that is weak enough to facilitate manual removal of the refill cartridge **106** by a user. For example, in some embodiments, a separation force between the magnet **126** and the refill cartridge **106** can range from 0.1 to 3 kilogram-force (kgf), such as about 0.3-1.2 kgf. In some embodiments, the magnet **126** can be a neodymium magnet, such as but not limited to an N**33** or N**38** magnet, or the like.

[0025] At the top of the inner shell **120**, an annular shoulder **128** can be disposed partially within the inner shell **120**, with an upper portion of the shoulder **128** extending axially from the inner shell **120** (and beyond the outer shell **118** as well). The shoulder **128** can be held in place in any suitable manner, such as by gluing or bonding or by friction such as press fit, detent, interlock, or the like. In some embodiments, an annular groove can be provided in the inner shell **120** configured to mate with a corresponding annular protrusion formed on the outer wall of the shoulder **128**. Pressing the shoulder **128** into the inner shell **120** elastically deforms at least one of the shoulder **128** or the inner shell **120** to allow the protrusion to move into the groove and secure the shoulder **128** in place. The shoulder **128** facilitates alignment of the cap **104** on the base **102**.

[0026] A base ring **130** may be disposed at a top end of the outer shell **118**, adjacent to the shoulder **128**. The base ring **130** provides a flat upper surface for mating with the cap **104**. The base ring **130** may include an inner lip extending downward from the upper surface. A magnet **116** can be disposed in the region bounded by the upper surface and inner lip of the base ring **130**, inner wall of the outer shell **118**, and upper surface of the inner shell **120**. The magnet **116** operates in conjunction with the magnet **114** of the cap **104** to facilitate securely retaining the cap **104** on the base **102**. The magnet **116** can be any suitable magnet having a magnetic strength that is strong enough to retain the cap **104** when disposed on the base **102** and that is weak enough to facilitate manual removal of the cap **104** by a user. For example, in some embodiments, a separation force between the cap **104** and the base **102** can range from 0.1 to 3 kilogram-force (kgf), such as about 0.3-2 kgf. In some embodiments, the magnet **116** can be a neodymium magnet, such as but not limited to an N33 or N38 magnet, or the like. In some embodiments, one of the magnet **114** or the magnet **116** can be replaced with a magnetic material.

[0027] As shown in FIG. **1** but also in greater detail in FIG. **2**, the refill cartridge **106** is configured to be movably disposed within the central opening of the base **102** (as indicated by arrows in FIG. **2**). For example, the refill cartridge **106** is configured to be placed into and retained in the central opening of the base **102**, and selectively covered or enclosed by the cap **104**, and can be manually removed from the base **102**. A gap **208** is disposed between the refill cartridge **106** and the base **102** to facilitate insertion and removal as well as relative rotation therebetween.

[0028] The refill cartridge **106** generally includes an outer shell **132** having an inner volume. The refill cartridge is configured to be retained within the base **102** by the magnetic field provided by the magnet **126**. For example, a bottom **134** of the refill cartridge **106** may be formed of a magnetic material, such as iron, steel, or the like. Alternatively, a magnetic material may be disposed above but proximate to the bottom **134** of the refill cartridge **106** such that the magnetic field of the magnet **126** provides suitable attractive force to retain the refill cartridge **106**. The bottom **134** of the refill cartridge can also provide a mass or weight to the refill cartridge **106** to advantageously provide enhanced feel and improved manual handling and alignment of the refill cartridge **106**.

[0029] The bottom **134** of the refill cartridge **106** is configured to interface with the drive post **124** of the base plug **122**. For example, the bottom **134** can include a recess **202** (as shown in FIG. **2**) having a size and shape to receive the drive post **124** with enough clearance to allow the bottom **134** of the refill cartridge **106** to rest on the bottom of the base **102** (e.g., on the magnet **126**). The drive post **124** and the recess **202** can generally include any suitable lock-and-key configuration. However, in some embodiments, the drive post **124** and the recess **202** have an octagonal shape. The inventors have discovered that the octagonal shape advantageously provides sufficient robust force transfer to the refill cartridge **106** while simultaneously facilitating ease of alignment of the refill cartridge **106** to the drive post **124** as compared to other shapes or configurations. For example, the inventors have discovered that, with the octagonal shape, the refill cartridge **106** can more easily align with, and in many cases to self-align with, the drive post **124**. The octagonal shape further is relatively easy to manufacture and provides a long lifetime by resisting deformation or breakage as compared to shapes having sharper corners or thinner portions.

[0030] As best depicted in FIG. **2**, in some embodiments, the entrance to the recess **202** can include a relief feature **204**, such as a radius, bevel, chamfer, or the like. In some embodiments, the relief feature **204** is a chamfer at an angle of about 30 to about 60 degrees. Alternatively or in combination, the drive post **124** can include a relief feature **206**, such as a radius, bevel, chamfer, or the like. In some embodiments, the relief feature **206** is a chamfer at an angle of about 35 to about 55 degrees.

[0031] In some embodiments, the bottom **134** of the refill cartridge **106** is a monolithic structure, such as the bottom **300** shown in FIG. **3**. Individual features of the bottom **300** can also individually be incorporated into the bottom **134** without requiring the combination or inclusion of all features described with respect to the bottom **300**. The bottom **300** includes an annular base **302** with a central axial protrusion **304**. In some embodiments, the protrusion **304** is circular/cylindrical. However the protrusion can have other shapes as discussed in more detail below with respect to the bottom **104**. A recess **306**, similar to the recess **202**, is disposed within the central axial protrusion **304**. A relief feature **310** is disposed about the opening of the recess **306** along a bottom surface **312** of the bottom **300**. A top edge **308** of the central axial protrusion **304** can also include a relief feature, such as a radius, bevel, chamfer, or the like. An outer lower edge **314** of the annular base **302** can also include a relief feature, such as a radius, bevel, chamfer, or the like. FIG. **4** depicts a bottom view of the monolithic bottom **300** more clearly showing the octagonal shape of the recess **306**. FIG. **5** depicts an isometric top view of the monolithic bottom **300** more clearly showing the configuration of the base **302** and protrusion **304**. FIG. **6** depicts an isometric bottom view of the refill cartridge **106** showing the outer shell **132** and the bottom **134** along with the recess **202** (or recess **306**) configured to mate with the drive post **124**.

[0032] Returning to FIG. **1**, in some embodiments, the refill cartridge **106** is configured to extend and retract a product based upon manual rotation of the refill cartridge **106** with respect to the base **102**. For example, in some embodiments, the refill cartridge **106** can include an elevator **138** to selectively extend or retract product **146**. The elevator **138** generally includes an outer shell **140**, an inner body **142**, and a holder **144**. A simplified description of one suitable elevator **138** is described below. However, other product position control mechanisms, including other elevators **138** as known in the art can be used. Alternatively, in embodiments where product extension and/or retraction is not desired or necessary, for example when configured to hold and/or dispense a fragrance or other liquid, the elevator **138** can be omitted.

[0033] The outer shell **140** of the elevator **138** is rotationally fixed to the outer shell **132** of the refill cartridge **106** and includes one or more spiral tracks disposed in sidewalls of the outer shell **140**.

[0034] The inner body **142** is disposed in and adjacent to the outer shell **140**. The inner body **142** is rotationally movable with respect to the outer shell **140** and includes one or more vertical tracks along a length of desired axial movement of the product **146**. The inner body **142** is rotationally movable with respect to the outer shell **132** of the refill cartridge **106** and is rotationally fixed with respect to the bottom **134** of the refill cartridge **106**. For example, the inner body **142** can be at least one of press fit, glued, pinned, threaded onto, or otherwise fixed to the bottom **134**. The bottom **134** can be rotationally movable with respect to the outer shell **132** of the refill cartridge **106**. The bottom **134** can extend slightly beyond the bottom of the outer shell **132** to prevent interference between the outer shell **132** and the magnet **126**.

[0035] For example, the bottom **134** includes a post **135** that fits into a corresponding opening in the inner body **142** of the elevator **138**. The post **135** can generally be any shape, such as square, triangular, hexagonal, circular, or the like. In non-circular embodiments, corners of the shape further facilitate prevention of rotation of the post **135** with respect to the inner body **142**. In some embodiments, the post **135** is glued in addition to having a friction fit. In some embodiments, one or more of the inner body **142** or the post **135** can include anti-rotation features (e.g., ribs, grooves, protrusions, or the like) to help ensure the inner body **142** and the bottom **134** remain rotationally locked and move as one solid entity.

[0036] A holder **144** is disposed in and adjacent to the inner body **142** and includes one or more radial protrusions that correspond to and fit within the one or more vertical tracks of the inner body **142**. The one or more radial protrusions further extend through the vertical tracks and into the one or more spiral tracks of the outer shell **140**. The one or more protrusions of the holder **144** rotationally fix the holder **144** with respect to the inner body **142** while allowing axial movement of the holder **144** along the one or more vertical tracks of the inner body **142**.

[0037] In operation, rotation of the bottom **134** of the refill cartridge **106** with respect to the outer shell **132** (for example, by holding the outer shell **132** and rotating the base **102**) will in turn rotate the inner body **142** and holder **144** with respect to the outer shell **140** of the elevator. During rotation, the holder **144** is displaced axially along the one or more vertical tracks in a direction dependent upon direction of rotation. Rotation in one direction will cause the holder to move closer to the top of the refill cartridge **106** and rotation in the other direction will cause the holder to move closer to the bottom of the refill cartridge.

[0038] FIG. **9** is an exploded schematic side view showing interrelation of major components of a cosmetic package having a refill system in accordance with embodiments of the present disclosure. For example, in some embodiments, a refill cartridge **106** can be configured to hold a solid product (as indicated by configuration **902**), such as product **146** disclosed above. In other embodiments, a refill cartridge **106** can be configured to hold a liquid product (as indicated by configuration **904**), such as but not limited to a fragrance, perfume, cologne, aftershave, deodorant, or the like. In the configuration **904** to hold a liquid product, the elevator function of the refill cartridge **106** is not needed and can be omitted.

[0039] While the foregoing is directed to embodiments of the present disclosure, other and further embodiments of the disclosure may be devised without departing from the basic scope thereof.

1. A cosmetic package, comprising:
- a base having an inner volume and a drive post disposed proximate a bottom of the base, the drive post extending into the inner volume;
- a refill cartridge configured to be removably disposed within the inner volume of the base, the refill cartridge including a recess configured to mate with the drive post; and
- a cap configured to rest on the base and enclose the refill cartridge within the inner volume.

2. The cosmetic package of claim **1**, wherein the drive post and the recess each have an octagonal shape.

3. The cosmetic package of claim **1**, wherein the drive post includes a relief feature along an upper edge of the drive post.

4. The cosmetic package of claim **3**, wherein the relief feature is a chamfer.

5. The cosmetic package of claim **3**, wherein the recess includes a relief feature along a lower edge of the recess.

6. The cosmetic package of claim **5**, wherein the drive post relief feature and the recess relief feature are each a chamfer.

7. The cosmetic package of claim **1**, wherein the recess includes a relief feature along a lower edge of the recess.

8. The cosmetic package of claim **7**, wherein the relief feature is a chamfer.

9. The cosmetic package of claim **1**, wherein the refill cartridge is magnetically retained in the base.

10. The cosmetic package of claim **9**, further comprising:
- a magnet disposed proximate the bottom of the base, wherein the refill cartridge has a bottom formed of a magnetic material.

11. The cosmetic package of claim **10**, wherein the bottom of the refill cartridge is steel.

12. The cosmetic package of claim **9**, wherein a separation force to separate the refill cartridge and the base when magnetically coupled together is 0.1 to 3 kgf.

13. The cosmetic package of claim **1**, wherein the cap and the base are configured to magnetically retain the cap on the base.

14. A cosmetic package, comprising:
- a base having an inner volume and a bottom plug disposed proximate a bottom of the base, the bottom plug including a drive post extending into the inner volume;
- an annular magnet disposed atop the bottom plug and having the drive post extending through the central opening of the annular magnet;
- a refill cartridge configured to be removably disposed within the inner volume of the base, the refill cartridge having a bottom formed of a magnetic material and including a recess configured to mate with the drive post; and
- a cap configured to rest on the base and enclose the refill cartridge within the inner volume.

15. The cosmetic package of claim **14**, wherein the cap and the base are configured to magnetically retain the cap on the base.

16. The cosmetic package of claim **14**, wherein the drive post and the recess each have an octagonal shape.

17. The cosmetic package of claim **14**, wherein at least one of the drive post or the recess includes a relief feature to facilitate alignment of the recess with the drive post.

18. The cosmetic package of claim **17**, wherein the relief feature is a chamfer.

19. The cosmetic package of claim **14**, wherein a separation force to separate the refill cartridge and the base when magnetically coupled together is 0.1 to 3 kgf.

20. A cosmetic package, comprising:
- a base having an inner volume and a bottom plug disposed proximate a bottom of the base, the bottom plug including a drive post extending into the inner volume, wherein the drive post is octagonal and includes a chamfer along an upper edge of the drive post;
- an annular magnet disposed atop the bottom plug and having the drive post extending through the central opening of the annular magnet;
- a refill cartridge configured to be removably disposed within the inner volume of the base, the refill cartridge having a bottom formed of a magnetic material and including an octagonal recess configured to mate with the drive post, the recess having a chamfer disposed along a lower edge of the recess to facilitate alignment of the recess with the drive post; and
- a cap configured to rest on the base and enclose the refill cartridge within the inner volume, wherein the cap and the base are configured to be magnetically retained to each other.

* * * * *